NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L22.2 Strata + Qwen3.8-Flash-Next local inference (Colab A100)

Group Idea Lab

Status Promoted

Period 2026-10-13 00:15 — 2026-10-21 00:51

Stage conclusion

Project report

Strata + Qwen3.8-Flash-Next local inference (Colab A100)

Objective

Benchmark Strata (MIT, pinned at v0.1.43) + Qwen3.8-Flash-Next (125B MoE / 6B active, unsloth UD-IQ4_XS ~93.7 GB) on Colab A100: whether it runs, its speed, cost, and how useful it is for us. Deliverables: feasibility and benchmark table, log paths, and a one-sentence conclusion (feasible / only feasible at a lower tier / not worth it).

Current status

Conclusion (10-13 01:20, raw logs verified by the lead): Feasible. On Colab A100 high-mem, Strata v0.1.43 + UD-IQ4_XS runs 32K without downgrading; for 32K input, time to first token is ~10 s, prefill is ~2,990 tok/s, and decode output is 35–59 tok/s; needle 3/3. The catch is that **every new Colab session requires ~40 minutes of setup (~4 CU)**, making it suitable for "batch-processing a set of long documents and then shutting down," rather than serving as an always-on service. This round cost ~5 CU (cap: 30).

Feasibility and benchmarks

Configuration	Runs?	Speed	Cost	Issues
Colab A100 high-mem hardware probe (10-13)	Yes	—	<0.1 CU	Must pass `--high-mem`
Strata v0.1.43 (commit) + unsloth UD-IQ4_XS (rev, 93.68 GB), 32K context	Yes, without downgrading	Peak VRAM 62.8 GiB/80; process RAM 7 GiB (weights backed by 159 GiB page cache)	—	No official prebuilt engine; requires on-the-fly compilation
Initial setup: weight download + compilation + pulling 5 GB MTP + packaging	Yes	~40 min (setup 1,796 s + weight verification)	~4 CU / per new session	Colab sessions do not persist disk, so every session starts from scratch; the script downloaded weights twice (disk reached 96%), use `--gguf-dir` next time
Engine load to ready	Yes	35 s	<0.1 CU	Default port 8080 is occupied by Colab; must change port (used 8090, bound to 127.0.0.1)
Short prompt (34–48 tok) ×3	Yes	First token 0.4–0.7 s; output 32–129 tok/s (median 36)	0.2 CU total	Output speed depends on MTP speculative hit rate and fluctuates widely
8K (~7,860 tok) ×3	Yes	First token 4.0–4.2 s; prefill ~1,970 tok/s; output 46–68 tok/s	Included above	
32K (~30,190 tok) ×3	Yes	First token 10.1 s; prefill ~2,990 tok/s; output 35–59 tok/s (median 43)	Included above	Author claims 60–160 tok/s; under long inputs we only reach the low end

Configuration	Runs?	Speed	Cost	Issues
needle 30.2K, 10%/50%/90%	3/3 correct	First token 10.1–10.5 s	Included above	Single-needle exact string retrieval only; does not represent long-context comprehension
Total			**~5 CU** (session ~00:20–01:13; Gemini reported 4.76–5.65)	01:15 the lead checked `colab ls`: no active sessions

License (original text verified by the lead from HF Qwen/Qwen3.8-Flash-Next LICENSE): MIT-style license requiring retention of the copyright notice; commercial products with >100 million MAU or >$20 million monthly revenue must display the model name in the UI; commercial "Model-as-a-Service / AI office or coding assistant" offerings require a separate license, but **internal use not open to third parties is exempt from this restriction**. There are no obstacles to internal use within the lab.

Attempts and results

Date	What was done	One-line result
10-13	Colab A100 high-mem hardware probe	80 GB VRAM / 167 GB RAM / 194 GB disk / CUDA 13.0
10-13	Benchmark by the gpu-gemini agent (reassigned from the ideas agent because of missing credentials)	Succeeded; 32K output 35–59 tok/s, needle 3/3, ~5 CU

Lead verification note: speed and needle numbers matched runner.log and server.log item by item. The lead corrected three points in the Gemini summary: (1) "35 s cold start" is only engine loading, whereas a new session actually takes ~40 min; (2) expert weights are mainly cached in VRAM (server.log: 55.43 GiB), not "entirely in system RAM"; (3) the 8K result from attempt #2 (first token 0.08 s) hit the cache and was discarded, and the final table uses the full run from attempt #3.

Usefulness to us and what not to do

- Useful for: One-off batch processing of long documents (standards/codes, long papers, full batches of logs, ≤32K tokens/doc), where data stays within Google and there is no per-token billing; can also serve as offline backup inference for bots. At ~4 CU startup cost + 5.4 CU/hour, it is only cost-effective when processing dozens of long documents in a single batch.
- Not suitable for: Always-on services (Colab sessions time out and disconnect after prolonged periods without output); long documents exceeding 32K (not tested in this round); replacing the platform's primary models.

- What not to do: Do not change the platform default model; bind the API only to 127.0.0.1 without opening public tunnels; do not use `--dangerously-skip-permissions`; do not provide inference services to third parties (License Clause 2).

Related projects

- Sub-project "full-text literature search + wind engineering data extraction" (originating from discussions in this node on 10-13) was promoted to the platform tool `pages` on 10-14; at 10-16 02:38 all ~556,000 pages across the three desktop sources were handed over, awaiting round5 blind testing after ingestion.
- This node only covers local LLM inference (Strata + Qwen and subsequent local model benchmarks). Evaluations and conclusions on using local LLMs for batch utility tasks are documented separately, with run.py and the cold-start numbers on this page serving as their starting point.

10-17 restart: Colab cold-start mitigation + G4 (PI, 10-17)

- After asking about "Colab's inability to stay always-on," the PI asked to put mitigation measures into practice and hand them to ComputeOps and all line leads once verified. Dispatched the ideas agent to compare caching options (small artifacts on the workstation / Google Drive / re-downloading) to see how much new-session setup time can be reduced from ~40 min; benchmark Colab G4 (while also testing whether Strata v0.1.45 is faster). ≤15 CU.
- Keep-alive utility is already available on the platform (handling idle reclamation and 24 h lifetime reminders) and is used directly.
- Google Drive: At 10-17 07:5x the lead tested `colab drivemount` (in a CPU session, since stopped). Every new Colab VM prompts for browser authorization (ephemeral credentials) rather than a one-time setup, so unattended bots cannot use it; instructed the ideas agent to switch to workstation small-artifact caching + Hugging Face re-downloading, and separately list "the portion that only Drive can save" for the PI's decision. The CLI OAuth token only has `drive.file` scope, and we did not attempt to pass the token into Colab (no credential export).
- **10-17 G4 benchmark (ideas, verified by Lead)**: G4 = RTX PRO 6000 Blackwell with 96 GB VRAM, 176 GB RAM, 194 GB disk, 48 cores (`--high-mem`; engine binary `strata-v0.1.45-sm120` cached, restoration not yet verified); large artifacts (weights, MTP, and packed bundle totaling ~100 GB) can only reside on Drive, pending `rclone` authorization.
- **Long-term Drive credentials**: Approved by the PI on 10-17; `rclone` restricted to `drive.file` scope (decision). The first authorization link at 08:13 was inadvertently invalidated by a lead connectivity check; at 10-17 10:58 the PI completed authorization (clicking the Google link on any device and returning the redirect URL to the lead); workstation (mode 600) remote `gdcache:` is active, Drive has 28.3 TiB free of 30 TiB, cache directory created and read/write tests passed; authorization logs deleted. Note: `drive.file` exposes all files created via the public `rclone` client (including a cluster archive folder from 10-14); the public `rclone` `client_id` will be retired within 2026 and must be replaced with a self-owned

`client_id` (to do). At 10-17 11:00 dispatched the ideas agent to store large artifacts on Drive + benchmark new-session recovery, ≤16 CU this round (project cumulative cap 25).

- **10-17 Drive large-artifact cache test (ideas)**: The public `rclone` `client_id` was rate-limited by Google: the 77 MB engine uploaded at 1.8 MB/s, the 1.5 GB packed bundle failed to finish in 20 min and hit `RATE_LIMIT_EXCEEDED`, and even downloads to the workstation were rejected. Only small files such as the engine and config were saved on Drive; new-session recovery was not tested. ~7.8 CU (project cumulative ~16.5/25). Conclusion: using Drive requires switching to a self-owned OAuth `client_id` created by the PI (requested); first dispatched the ideas agent to test how much time "caching only the engine without Drive" saves (next order, ≤7 CU).
- **10-17 Engine-only cache test (ideas, verified by Lead)**: Caching only the 77 MB executable is insufficient: Strata's `setup.py` must also see `BUILD.json` (recording the source hash and GPU architecture) to skip compilation; our workstation cache lacked this file and therefore recompiled anyway, saving 0 minutes. Also found that v0.1.45 requires `--gguf-dir` to point to already-downloaded weights, otherwise it downloads them again. ~3.3 CU (project cumulative ~19.8/25).
- **Next step (set by the lead)**: Wait for the PI to create a self-owned OAuth `client_id` and then run one final two-session round: Session A performs a complete setup once, saving the entire engine directory (including `BUILD.json`) to the workstation and the packed bundle and MTP to Drive; Session B boots fresh to restore only, measure setup time, and run a short prompt and 32K test. Estimated ~8 CU, raising the project cap to 30 CU at that time. No Colab sessions will be opened before then.
- **10-17 17:0x Switched to the PI's existing authorized channel**: Copied the desktop `rclone` remote `gdrive` (the PI's self-owned OAuth client with `drive.file` scope, the same one used for CRC archiving) to the workstation (mode 600); workstation tests measured 9.6 MB/s upload and 31.5 MB/s download with no rate limiting. Deleted the old `gdcache:` remote (rate-limited public client) and cleared its test cache (engine copy, config, and tokenizer files, all reconstructible while the original engine remains on the workstation). The PI no longer needs to create a new `client_id`. Dispatched the ideas agent for the final round (Session A full setup and cache population, Session B fresh recovery benchmark), ≤10 CU, raising the project cap to 30.
- **10-17 final round (ideas agent, lead verified raw setup log lines and stop records)**: Cached the complete engine directory (3 files, 91.6 MB including BUILD.json, stored on both workstation and Drive) + 1.5 GB packed bundle + 6.9 GB MTP (on Drive, 8.5 GB total); weights re-downloaded from HF as usual. Fresh G4 session: Drive fetch 142 s, HF weights 384 s, setup 40 s (compilation and packing skipped), engine load 150 s, cutting comparable stages from 1,907 s to 716 s (~32→12 min) and allocation-to-ready time to 14 min 15 s; 1 short prompt and 1 30K long-context test both succeeded. Colab↔Drive transfer reached 37–58 MB/s upload and 60–84 MB/s download. ~7.5 CU this round, ~27.3/30 CU project cumulative. Handoff sent to Platform Lead and ComputeOps: recorded by ComputeOps on 10-17; platform committed to merging into the shared documentation (re-checked at 10-18 18:07: README is still the 10-17 18:12 version, already containing the gdrive section and "do not re-authorize on either side"; workstation and desktop refresh token fingerprints remain identical (not yet separated), and Drive access is normal; verified in place by the lead at 10-17 22:20: README already includes a gdrive: caching section and credential rules; workstation rclone.conf was updated by the platform at 18:10, remaining mode 600 with gdrive as the sole remote), and the workstation rclone.conf is now managed by the platform (path and permissions unchanged), which will issue a separate token for the workstation so the workstation and desktop do not invalidate each other.

- **10-21 summary and future report**: L32.0 Stage 3c (10-18) selected Qwen3.8 as the v0 base model.
- **10-18 Promoted**: Approved by the PI to merge into L32 AI Teaching Lab · L32.0 in-house teaching large language model; handoff sent and received on 10-18: incorporated into L32.0 and scheduled for Stage 3c (launching Strata from cache on G4 to benchmark `eval_v2`/`eval_v2h` on the same basis as `gpt-oss-120b`, ≤10 CU, reaching a same-day conclusion on the v0 base model), adopting caching + keep-alive as the default, with teaching demonstrations pending the PI's decision on Stages 4/5. L22 will no longer dispatch new tasks here, and materials are retained in place for read-only reference.
- 10-18 the PI: This should be very useful for subsequent teaching experiments, and its use cases and home should be considered. The Lead drafted a promotion proposal (recommending merging into L32.0 in-house teaching large language model), pending the PI's approval. Colab balance: 2,222 CU (checked on 10-18).
- The PI has an RTX 5090 32 GB available (10-17); system RAM size pending inquiry.

Boundaries

- Run only on Colab and do not download weights onto the workstation; do not change the platform default model; bind the API only to 127.0.0.1; do not use `--dangerously-skip-permissions`; do not save any tokens.
- Verify the original text of the Qwen Community License 1.0; run weight downloads in the background with staged checks to prevent disconnection; must stop the instance when finished.

Open items and next steps

- **Reason for On hold status**: The feasibility question has been answered (feasible); there is currently no concrete batch of long-document tasks that requires it, so maintaining it standalone is not cost-effective.
- **What is missing**: A practical use case (for example, a batch of ≤32K codes/papers requiring batch extraction that should not be sent to external APIs).
- **When to restart**: When such a use case arises, reuse the existing setup (switching to `--gguf-dir` to avoid downloading weights twice), budgeting ~4 CU startup + 5.4 CU/h; if >32K is needed, add a 64K test tier.
- The ideas credential issue has been confirmed fixed by the platform (normal since 10-13 06:09, per the Platform Lead); this task was already completed by gpu-gemini and will not be reassigned.
- Not sent for review: This is an engineering feasibility benchmark rather than a scientific conclusion; the numbers have been verified item by item against the raw logs by the lead.

Literature basis

- (The author's speed table and needle test had not been independently replicated; MODELS.md places UD-IQ4_XS in the 96 GB+ RAM tier.)

PI notes

Key figures